

Software Acceleration Team (SwAT)

intro leadership/formation meeting

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@mere-solace on github

Contents



1	Who am I	4
1.a	Background	5
2	Why make a club?	6
2.a	Community	7
2.b	Classes	8
2.c	Interviews and Job Hunting	9
2.d	What's missing	10
3	What is SwAT	11
3.a	In a nutshell	12
4	Goals	13
4.a	Organize	14
4.b	Develop	15
4.c	Educate	16
4.d	Educate - Pace	17
4.e	Connect	18
5	Specifics	19

Contents (ii)



5.a Project lifecycle	20
5.b Defended Review	21
5.c Membership requirements	22
6 Housekeeping	23
6.a Officer selection	24
6.b Open items	25
6.c Weekly meetings	26
6.d Next steps	27
6.e The End	28

1 Who am I

Background

- I am a senior Computer Science and Math major with a Discrete Math and Operations Research concentration.
- Built a large-scale combat simulation plugin in minecraft using java + paper, explored ML and quant workflows, and created a 16-bit CPU simulation in multisim.
- Was involved with Student Managed Investment Fund: This is really where I got the idea for this club, as well as the defended review/presentation format.
- Why I am starting this instead of joining something that is already established?

2 Why make a club?

The incredible competition performance is just one manifestation of MIT's vibrant community of students who love doing math and discussing math with each other, students who through their hard work in this environment excel in ways beyond competitions, too.

— MIT News — February 28, 2025

My takeaway

High performance is a byproduct of community. Gather people with the same goal together, talking and building, (+ structure and hard work) and the results follow.

Classes



- Most CS and software courses at KSU focused on theory and general skills/knowledge
- Little room for practical application
- Tools and frameworks are deliberately de-emphasized
- Lowerclassmen never touch real systems

This is not meant to fault the department. The initiative falls to the individual student, which can be a good thing, but a standard question I hear is: “what should I be working on.”

Interviews and Job Hunting



Interviewers ask about collaborative work explicitly, or they weight it implicitly over solo work.

Being able to answer and explain the following is important in interviews:

- Tell us about a time you worked with others on software
- What was the problem/goal, and how did you get there
- Why did you build it at all
- What tools did you use, and can you explain them
- What did you learn
- What was the outcome

Theory+DSA+‘I can learn quickly’?

3 What is SwAT

In a nutshell

Many wonder where the door into cs/tech jobs is now. This is the door: my aim is to provide students with real experience that they can speak to without having to be hired, so that they **can** get hired.

Mission

A space for computing students (CS, SWE, cybersecurity, game design, math) to get practical engineering experience beyond the classroom.

Organize

Develop

Educate

Connect

From Article II of the constitution

4 Goals

Self-study is necessary, and solo work is meaningful; however, progress is better with support and guidance.

- Long-form team projects with deadlines and expectations.
- version control, code review, DevOps, documentation, and non-trivial tools

Strengthen fundamentals through **project presentations with defended technical reviews.**

- The team presents the project
- A panel will question your language, architecture, tooling, and vc
- You defend the decisions, or you learn why you cannot, before you get into an interview with the project

CCSE's stance:

- Frameworks change; concepts do not
- A student can pick up a framework quickly
- => teach the theory

This club can fill that gap:

- You can't ship a project without frameworks and tools
- Avoiding them because "they might change" can prevent you from doing, and rob you of the confidence that comes with doing
- We will teach and use them here, with member-led sessions on tools/frameworks

Educate - Pace



- CSE 1321 and 1322 take a long time
- For most students that pace is correct
- For students who want more, can handle more, and are aiming at hard fields, it is too slow
- We will bring lowerclassmen up to speed faster, on their own initiative
- Upperclassmen benefit as well

Eventually:

- Industry partnerships and guest reviewers
- Client work under a written agreement
- Support for senior projects and students in SWE courses
- Alumni returning to sit on review panels & mentor students

5 Specifics

Project lifecycle



Pitch

Scope

Build

Defended Review

Retrospective

- Hard deadlines (the main defense against scope creep)
- Room to change direction, or to halt a project that is not working
- Technical decisions get written down as they are made
- Branches for features and bugs; version control history can serve as the record of the project

Defended Review



- This is a learning experience, a mock interview, if you will
- A panel questions the team against a written rubric
- Then the room gets to ask anything
- Assessed on: code quality, repo and version control practice, architecture, tool choice, documentation, results, and whether you can explain any of it

Membership requirements



Per semester, to count as an active member:

- Attend 50% of general meetings
- and either:
 - contribute to a project team or
 - deliver one presentation
- No dues

6 Housekeeping

Officer selection



Required by Owl Life:

- President
- Vice President / Reservation Delegate
- Treasurer

Also on the Executive Board:

- Secretary
- Director of Projects
- Director of Outreach

Role descriptions: **Article V, Section 1** of the constitution. President and Treasurer cannot be the same person, and the board needs 4 distinct students.

Open items



1. **Officer eligibility.** Article V, Section 2 requires one semester of active membership before running. We need a founding exception on the record.
2. **“Good standing” is still TBD** in the bylaws. Constraint from Dmitri: it cannot be inferable from data a tutor or student employee can see.

Which evening?

- Day of week
- Start time and length
- Room - VP will reserve through Owl Life (at least 48 hrs beforehand)

Next steps



- GitHub access
- Spruce up the Discord; make a teams team
- Update constitution?
- Register the club and then make an Owl Life page
- Logo?
- Officers must complete required training.
- Two presentations and two project ideas ready to go before the first open meeting

The End



Thank you

Questions, comments, suggestions, ideas, and pushback are all welcome.